

G-Q: Determine whether or not there is a cycle in an undirected graph. In an undirected graph..., the nodes are numbered from 0 to 88, and the edges are: (0, 73) (0, 51) (0, 10) (0, 63) (0, 28) (1, 62) (1, 57) (1, 84) (1, 61) (1, 5) (1, 24) (2, 84) (2, 3) (2, 66) (2, 68) (2, 17) (2, 35) (2, 34) (2, 15) (3, 39) (3, 52) (3, 16) (3, 15) (3, 8) (4, 69) (4, 85) (4, 36) (4, 72) (5, 44) (6, 77) (6, 7) (7, 85) (8, 64) (8, 23) (8, 28) (9, 34) (9, 31) (9, 61) (9, 28) (10, 26) (11, 37) (11, 39) (11, 19) (11, 64) (13, 73) (13, 61) (13, 80) (13, 85) (14, 86) (14, 59) (14, 32) (14, 58) (14, 85) (14, 66) (15, 43) (15, 48) (15, 73) (15, 19) (15, 47) (15, 68) (16, 46) (16, 60) (16, 84) (17, 44) (17, 72) (17, 36) (17, 37) (17, 61) (18, 20) (18, 24) (18, 22) (18, 41) (19, 45) (19, 83) (20, 25) (20, 29) (21, 38) (21, 64) (21, 24) (21, 22) (21, 34) (22, 23) (23, 34) (23, 30) (23, 83) (24, 47) (24, 50) (25, 59) (25, 42) (25, 70) (25, 72) (26, 45) (26, 30) (26, 87) (26, 80) (26, 50) (27, 77) (27, 58) (27, 60) (27, 29) (28, 36) (28, 59) (28, 88) (29, 31) (29, 33) (29, 56) (29, 55) (29, 78) (29, 73) (29, 82) (30, 34) (31, 58) (31, 70) (32, 78) (33, 73) (33, 68) (34, 48) (34, 88) (34, 51) (34, 66) (34, 54) (35, 43) (35, 56) (36, 44) (36, 39) (37, 57) (37, 71) (38, 47) (38, 58) (38, 88) (39, 74) (39, 48) (39, 45) (39, 42) (40, 60) (40, 59) (40, 71) (40, 76) (41, 79) (42, 60) (42, 62) (42, 72) (43, 57) (44, 47) (44, 74) (44, 75) (44, 82) (44, 70) (45, 68) (45, 67) (45, 71) (46, 84) (47, 56) (48, 75) (49, 71) (49, 58) (50, 76) (51, 55) (51, 87) (51, 79) (52, 78) (56, 69) (56, 83) (56, 79) (57, 76) (57, 70) (57, 72) (57, 78) (59, 70) (60, 80) (60, 76) (60, 77) (60, 75) (61, 74) (61, 77) (62, 84) (62, 66) (66, 69) (68, 88) (70, 88) (70, 76) (71, 81) (71, 72) (71, 86) (72, 75) (73, 82) (73, 83) (74, 85). **Is there a cycle in this graph?**

GPT-4: No, there is no cycle in this graph. ✖

GraphWiz: There are many possible ways to find a cycle in this graph, but one example is: Starting from node 0, we can go to node 73 (via edge 0-73), then to node 82 (via edge 73-82), then to node 44 (via edge 82-44), then to node 17 (via edge 44-17), then to node 36 (via edge 17-36), then to node 28 (via edge 36-28), and back to node 0 (via edge 28-0). This forms a cycle [0-73-82-44-17-36-28-0] without revisiting any edge. So, there is a cycle in this graph. ✔